

$^{32}\text{S}(^3\text{He},\text{p})$ 1977Bi04

$^{32}\text{S}+1\text{p}+1\text{n}\rightarrow^{34}\text{Cl}$ from $J^\pi=0^+$ ^{32}S ground state.

1977Bi04,1976Bi14: an 18-MeV ^3He beam was provided from the University of Pennsylvania tandem accelerator. The target was gaseous H_2S . Protons were detected on nuclear emulsions in a multi-angle broad range magnetic spectrograph with an energy resolution of 22-25 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis.

1976De24: a 24.5-MeV ^3He beam was produced from the Princeton azimuthally varying field cyclotron. The target was CdS with natural ^{32}S . Reaction products were detected using a freon-cooled $\Delta\text{E-E}$ Si detector telescope with FWHM=40 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured $^{32}\text{S}(\alpha,\text{d})^{34}\text{Cl}$.

1972Na23: a 24-MeV ^3He beam was produced from the MP tandem accelerator of the Max Planck Institut fur Kernphysik in Heidelberg. Targets were PbS on carbon backings. The reaction proons were momentum-analyzed by a multigap magnetic spectrograph and detected using 50- μm Ilford L4 nuclear emulsions with a resolution of 40 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from zero-range DWUCK-DWBA analysis.

Others: 1991Ru05 measured cross sections for $^{32}\text{S}(^3\text{He},\text{p})^{34\text{m}}\text{Cl}$, $E=4.0\text{-}12.0$ MeV. 2017Ay03 measured $^{32}\text{S}(^3\text{He},\text{p})^{34}\text{Cl}$ as a byproduct of $^{24}\text{Mg}(^3\text{He},\text{p})^{26}\text{Al}$.

 ^{34}Cl Levels

E(level) [†]	L [†]	Comments
0.0 25	0	L: 0 also from 1972Na23.
145.3 16	2+4	
461.8 34	2	L: other: L=2(+0) from 1972Na23.
664.7 54	0	L: other: L=0(+2) from 1972Na23.
1230.4 73	(2)	L: other: L=2 from 1972Na23.
1880.0 53	2	
2163.0 87	2	E(level): doublet.
2376.8 75	(4)	
2583.0 41	0	L: 0 also from 1972Na23.
2614.2 38	(2)+(4)	
2720.5 39	1+3	
3127.0 27	0	L: other: L=0+2 in 1972Na23.
3336.4 31	2	L: 2 also from 1972Na23.
3383.2 43	2	
3548.4 52	3	
3601.1 49	(3,4)	
3635.2 48	5	
3804? 29		
3982.6 72	(0)+3	E(level): doublet.
4076.5 58	5(+3)	
4141.1 79	1+3	
4354.5 20	1	
4412.1 31	3	
4455.8 40	3	
4514.5 32	1+3	
4611.9 70	1+3	E(level): doublet.
4693.0 34	2,1+3	E(level): other: 4674 (1976De24).
4824.8 24	3,4	E(level): other: 4790 (1976De24).
4965.7 71	(1,2)	E(level): doublet.
5292		E(level): from 1976De24.
6319		E(level): from 1976De24.
6724		E(level): from 1976De24.

[†] From 1977Bi04, unless otherwise noted.